

BridgeTrafficLoadSim:

*Long Run Simulation Model
for Bridge Loading*

Version 1.3.8



User Manual

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Acknowledgements

This program is based on three separate programs developed as part of the author's PhD research from 2001. In turn, these were based on work by Dr Samuel Grave, former PhD student at Trinity College Dublin. The current program which encompasses the functions of two of the previous programs has evolved since 2007 and has been much influenced by the work of Dr Bernard Enright, DIT.

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1 Introduction

1.1 The BridgeTrafficLoadSim Program

BridgeTrafficLoadSim will be referred to as BTLs hereafter.

BTLs generates artificial traffic and passes it across bridges determining various load effects. The traffic is generated according to a relatively simple model. There are several built-in influence lines for various load effects, but the user can input their own influence line also. The program outputs various quantities of interest, which are controllable by the user.

These programs have been in use in DIT and University College Dublin over a 10-year period. There have been multiple users, and so exhibit a fair degree of maturity. That is, the user should not get unexpected results when used correctly. Whilst the routines have been thoroughly tested many times, with ongoing changes, it is good practice to satisfy oneself as to the accuracy of the programs. Various forms of output should assist with this process.

1.2 The User Manual

Purpose

This User Manual has been written to explain the use of the BTLs program, and to explain its capabilities and limitations.

Notices

Points of significant importance are denoted as:

Important!

Typically, failure to adhere to these points will result in unexpected behaviour or a program crash.

Glossary

Load Effect The result of a calculation using any influence line. Total load on the bridge is sometimes referred to as a load effect therefore.

1.3 Program Release History

Version	Date	Description
1.0.0	5/7/12	Initial release to international users
1.0.1	21/7/12	- Added version number on screen output - Fixed problem with AllEvents output – it now outputs the last unfilled buffer properly.
1.0.2	24/7/12	Added a FatigueEvents output file type with max and min values of loading events in it. Only output if AllEvents is output - temporarily
1.0.3	27/7/12	- Bug fix: truck departures not always correctly calculated - fixed using 1e300 for timeOff variables. - Bug fix: reading single lane vehicle files caused crash - fixed. - Minor console output changes for more user information
1.0.4	13/8/12	- Console output for missing files. - Bug fix: reading multiple lane vehicle files crashed following v1.0.3 fix for single lanes!
1.0.5	27/8/12	Separate discrete influence lines now possible for each lane (IL option 2 in bridge definition file).

1.1.0	27/10/12	• Peaks over Threshold output • Basic statistics output • Created inheritance structure for output types • Flow Data statistics output • Restructured the BTLSin file • Fixed some bugs, especially one on flow generation
1.2.0	28/11/12	• Renamed output files for consistency • Added Influence Surface (IS) calculations • Amended BTLSin file structure accordingly • Added new file structure – DITIS for ISs, including wheel track width for each axle • Added POT counter file output and input specs in BTLSin • Program can now run without IL or IS files being present in folder, once not required. • Added option for vehicle output file format • Traffic folder location now can be specified in BTLSin • Updated generation of tri-modal-normal distributions to include deterministic and single-generation value for axle track widths • Added transverse position in lane variability through BTLSin • Added transverse axle track width generation file (ATW.csv) input and default value options
1.2.1	20/12/12	• Bug fix: influence surface calculations did not take transverse position into account – now fixed. • Bug fix: crash caused by traffic files with no vehicles in a lane is fixed.
1.2.2	27/3/2014	• Bug fix: VehicleBuffer.h(31) now max axles 20
1.2.3	14/6/2019	• Improved feedback on incorrect load effect definitions • TH file time precision is now 3 places • Bridge/IS files now can include comment lines ‘\\’
1.2.4	12/8/2019	• Added number of events of each truck count to statistics output
1.2.5	21/08/19	• Added MON file format • Cleaned up all warnings • Corrected minor bugs in flow/sim time calculations
1.2.6	25/08/19	• Generic classification of vehicles • Improved warnings • Clarified local and global lane numberings • Block numbering fixed to relative sim time • Correct MON file bug
1.3.0	28/08/19	• Generic flow and vehicle generation routines • Flow and vehicle models can now be mixed • Added new garage vehicle generator
1.3.1	30/08/19	• Minor bug fixes (kernels & RNG seed) • Started OpenMP, but crashes in x64 release • Bug fix: RNG seed, and kernels • Converted to shared_ptr • Still has memory leaks in CConfigData
1.3.2	2/09/19	• Fixed CVehicleBuffer bug, storing every vehicle forever. • Moved input files to Working directory • clarified memory leak reports • improved smart pointer usage
1.3.3	2/09/19	• Corrected overlap bug, and improved reporting of potential overlaps
1.3.4	3/09/19	• Fixed bug in MinGap calculations • Necessary to re-order sequence of start up

1.3.5	17/09/19	• RNG updated to C++11 STL • Updated Project Configurations: changed exe to x64 default, and now x86 is named x86 • More warnings quashed
1.3.6	20/09/23	• Minor performance tweaks • Use C++17 for the path lib • Support rainflow process • Many minor bug fixes for cross-platform compiling • Add two built-in load effects
1.3.7	4/7/26	• Fix some typos • Added the SiWIM CSV traffic input file format (format code 5, read only) • Rainflow counting rewritten to the ASTM E1049-85 four-point method; cycle counts are no longer dependent on output buffer size • Statistics output fixes: interval column alignment, added Min/Max columns, and trailing quiet blocks/intervals are now written up to the simulated end time • Bug fix: lanes with a zero-flow hour are no longer permanently silenced for the rest of the simulation • Approximately 1.8× faster single-core execution through I/O and copy improvements (bit-exact outputs) • If <code>BTLSin.txt</code> cannot be opened the program now reports an error and exits with a non-zero code, instead of proceeding with default values • Removed the Windows-only “PAUSE” console prompts; the program exits cleanly on all platforms and returns a proper exit code
1.3.8	7/9/26	• The console banner now reports the actual build version • Bug fix (changes results): the Grave vehicle model drew the GVW of trucks in direction-1 lanes from the direction-2 distribution, because the lane direction was never passed to the generator • Bug fix (changes results): the congested headway model used the gap coefficient of variation (Line 12) as an absolute standard deviation in seconds; it is again relative to the mean congested gap • Bug fix: the nominal (constant) vehicle model applied the axle-spacing and axle-weight coefficients of variation to each other’s quantity • A run with load effects enabled but an empty bridge file no longer sets the no-overlap length to 0 m • The no-overlap length (<code>m_MaxBridgeLength</code>) is now documented in the source

Installing the Program

BTLS does not require installation, it is a standalone executable program. Historically it was tested on Windows XP, Vista, and Windows 7; since v1.3.6 the program builds from source with CMake (C++17) on Windows, Linux, and macOS.

BTLS can be run in a 64 bit version, which is more efficient than the 32 bit version. The program is a single-threaded application and so cannot take advantage of multi-core processors. Therefore for maximum speed, prefer a computer with a fast single processor over a computer with a multi-core slower processor.

The program does not require much memory because:

- Only a small amount of input information is held in memory;

- Traffic is generated in 1-day blocks on a rolling basis;
- Output to file is made according to user input: this balances accessing the hard drive (which is slow) and memory requirements. Prefer to use memory than output to the hard drive often.

A computer with 1 GB of RAM is sufficient and other programs can continue to operate successfully.

BTLS uses two folders to operate:

- **Working folder:** the current folder in which configuration files and the executable exist;
- **Traffic folder:** a folder on the computer in which the traffic characteristics for sites are stored.

2 About BTLS

2.1 Introduction

BTLS performs efficient calculations of static traffic actions on bridges. It can generate artificial traffic and write it to file. It can calculate load effects from traffic files. However such simulations are limited by the file size that can be held in the computer memory. In an alternate mode, it can generate traffic and determine load effects simultaneously without recourse outputting traffic to file. In this mode the program can simulate 100s of years of load effect data quite quickly. The exact speed depends on many parameters, but in the worst case, 1000 years has been simulated in under 20 hours. For more typical cases 100 years takes about 1 hour to simulate.

BTLS is provided in two versions:

- BridgeTrafficLoadSim.exe: the 32-bit version.
- BridgeTrafficLoadSim_x64.exe: the 64-bit version, found to run faster than 32-bit version on 64-bit machines.

2.2 BTLS: Capabilities and Limitations

Capabilities

BTLS is able to:

- Generate artificial traffic from the traffic model of Caprani (2005 & 2012).
- Read in traffic and pass it over influence lines and surfaces;
- Use lane factors to account for lateral distribution of load effect due to transverse stiffness of the bridge;
- Use user-defined influence lines;
- Use separate influence lines for each lane of traffic (user-defined or in-built);
- Determine static load effects from generated or read-in traffic passing over defined bridges and either user-defined or built-in influence lines;
- Model one or two directions at the same time, with any number of lanes in each direction;
- Output different types of data for debugging and further analysis, as specified by the user.
- Output a file suitable for further fatigue analysis.
- Output data for block maxima or peaks over threshold approaches.
- Output traffic flow statistics and load effect statistics

Future plans include:

- Built-in fatigue calculation;

- Improved traffic model for greater generality;
- Possibly a visual user input interface.

Limitations

BTLS is not able to:

- Generate trucks with more than 5 axles;
- Determine the number of lanes or directions of traffic in the specified input traffic file, in advance of a simulation;
- Determine the input traffic file format;
- Determine dynamic load effects;
- Perform extrapolations for return periods.

Note that cars are assumed to be 4 m long and have GVW of 2 tonnes evenly distributed to each axle.

3 BTLS Input

3.1 Introduction

BTLS operates in one of three modes, which are numbered:

1. ***Gen & Sim***: In this mode traffic is generated in the program and simulated crossing the defined bridges;
2. ***Gen***: In this mode traffic is generated and output to file;
3. ***Read & Sim***: In this mode traffic is read from a file and simulated crossing the defined bridges.

Different types of input are required:

1. Traffic model files: for the generation of artificial random traffic;
2. Configuration file: the main user input file which configures each run of the program;
3. Supporting files: define bridges, influence lines and traffic flows to be used in the simulation.

Thus for a successful run the files required are:

Location	Files
Traffic folder defined in BTLSin.txt, usually C:\Traffic\[The Site] Working Folder (anywhere on computer)	Several – see Traffic Data input files section BridgeTrafficLoadSim.exe (or 64 bit version) BTLSin.txt Lane flow definition file Bridge definition file Influence line definition file Influence Surface definition file

Depending on the program mode and input values, BTLS will run if some of the above files are missing or not accessible. Warnings are usually given.

3.2 Traffic Files

The Grave Traffic Model (Model 0)

The model describing the physical characteristics of the traffic is defined in a series of files located in a folder, named after the site which is located, which is a sub-folder to the Traffic Folder. As of v1.2.0 this location can be specified in BTLSin.txt, but it is recommended to keep it at C:\Traffic\.

Important!

The traffic folder must reside at the location specified in BTLSin.txt (usually C:\Traffic\).

The traffic model is described by Caprani (2005), and is based on Grave (2001). Presently, 13 sites have been modelled accordingly and are indexed by BTLs as follows, and are located in sub-folders of the site name below:

Site Indices

Index	Site
1	Angers
2	Auxerre
3	A196
4	B224
5	A296
6	SAMARIS\D1
7	SAMARIS\D2
8	SAMARIS\D3
9	SAMARIS\S1
10	SAMARIS\S2
11	SAMARIS\S3
12	SAMARIS\D
13	SAMARIS\S

Auxerre is particularly important as the Eurocode load model LM1 was initially calibrated upon this traffic.

Traffic Data Input Files

The files then placed in this folder are of type comma separated values (*.csv). These file types are easily created in a spread sheet program, but can also be read or edited in a text editor.

Many of the vehicles properties are modelled with a three-mode normal distribution; that is, the data may be multi-modally normally distributed. There are three parameters required for each of the modes: the weight, ρ ; the mean, μ and the standard deviation, σ . The maximum number of modes allowed for is three; hence the 3×3 tabular format of the data. The units of the data are as per the traffic file convention explained in the Output section.

Important!

The files names must be as given for each modelled property.

The input defining the traffic flow and composition is made in the working folder as the executable.

Important!

The current traffic model only accounts for vehicles with up to 5 axles.

Note:

In the following, tri-modal normal distributions are often used to model parameters. Should deterministic values be needed instead, the distribution should have only one mode, set the mean to the required value, and set the standard deviation to zero.

Axle Spacing Definition

Asall.csv

This file stores the axle spacing data for all classes of trucks measured at the site. The values must be separated by commas. An example is:

```
1,50.7,3.7,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0
0.65,34.1,6.9,1,11.5,1.7,0,0,0,0,0
0.268,34,1.5,0,0,0,0,0,0,0,0
0.082,61.5,6,0,0,0,0,0,0,0,0
0.672,30.6,1.5,0.153,34.7,3,0.317,11.8,0.6,0,0,0
0.328,30.2,3.9,0.386,54.8,8.6,0.598,12.1,1.7,0,0,0
0,0,0,0.461,59.5,3.4,0.085,18.3,0.9,0,0,0
0.041,23.2,1.4,0.133,42,5.6,1,10.9,1.7,1,11,1.7
0.959,30.4,1.8,0.867,51.2,3.4,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0
```

This data may be more easily understood viewed in tabular form. The meaning of the rows and columns is also shown in relation to the ti-mode normal distribution adopted.

Class	Line	Spacing 1-2			Spacing 2-3			Spacing 3-4			Spacing 4-5		
		ρ	μ	σ	ρ	μ	σ	ρ	μ	σ	ρ	μ	σ
2-Axle	1	1	50.7	3.7	0	0	0	0	0	0	0	0	0
	2	0	0	0	0	0	0	0	0	0	0	0	0
	3	0	0	0	0	0	0	0	0	0	0	0	0
	4												
3-Axle	5	0.65	34.1	6.9	1	11.5	1.7	0	0	0	0	0	0
	6	0.268	34	1.5	0	0	0	0	0	0	0	0	0
	7	0.082	61.5	6	0	0	0	0	0	0	0	0	0
	8												
4-Axle	9	0.672	30.6	1.5	0.153	34.7	3	0.317	11.8	0.6	0	0	0
	10	0.328	30.2	3.9	0.386	54.8	8.6	0.598	12.1	1.7	0	0	0
	11	0	0	0	0.461	59.5	3.4	0.085	18.3	0.9	0	0	0
	12												
5-Axle	13	0.041	23.2	1.4	0.133	42	5.6	1	10.9	1.7	1	11	1.7
	14	0.959	30.4	1.8	0.867	51.2	3.4	0	0	0	0	0	0
	15	0	0	0	0	0	0	0	0	0	0	0	0

Axle Track Width

ATW.csv

This file is required when generating vehicles for use with an influence surface. It specifies the width from centre of pressure of the wheels on one side of the axle, to the other. In general this dimension is different to that of vehicle width, since the tyre width and number of tyres must be accounted for. This file is optional, and if it is missing then a default value specified in BTLSin.txt will be used instead.

This file again uses tri-modal normal distributions to model the track width of each axle for each class of vehicle. The distributions are specified using 3×3 matrices as is done for Axle Spacings. The file structure is then:

Class	Line Nos.	Axle 1	Axle 2	Axle 3	Axle 4	Axle 5
2-axle	1-3 4	3×3	3×3	N/A	N/A	N/A
3-axle	5-7 8	3×3	3×3	3×3	N/A	N/A
4-axle	9-11 12	3×3	3×3	3×3	3×3	N/A
5-axle	13-15	3×3	3×3	3×3	3×3	3×3

An example file is shown below which illustrates three separate model types for the track width:

1. A typical implementation using tri-modal distributions for each axle track width is shown for the 2-axle truck type. Notice the zero values for the redundant axles 3, 4, and 5.
2. Single values of track width can be specified for all the axles by only defining a distribution for the first axle, and leaving subsequent values all zero. This is done for 3-axle trucks in the example below.
3. Deterministic values can be defined as noted previously by specifying only one model, with mean of the require value and standard deviation of zero. This is done in the example file below for 4- and 5-axle trucks, showing the different track widths for the different axle types (steering, drive, trailer with dual tyres).

The track widths are specified in units of **centimetre**.

```

0.109,188,27.3,0.109,188,27.3,0,0,0,0,0,0,0,0
0.847,193.9,6.4,0.847,193.9,6.4,0,0,0,0,0,0,0,0
0.045,167.6,6.1,0.045,167.6,6.1,0,0,0,0,0,0,0,0
0.109,188,27.3,0,0,0,0,0,0,0,0,0,0,0
0.847,193.9,6.4,0,0,0,0,0,0,0,0,0,0,0
0.045,167.6,6.1,0,0,0,0,0,0,0,0,0,0,0
1,196,0,1,184,0,1,199,0,1,199,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,196,0,1,184,0,1,199,0,1,199,0,1,199,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0

```

Axle Weights

Aw2&3.csv

Two files are used, one for 2 and 3 axle trucks, the other for 4 and 5 axle trucks. This file contains the axle weight information for the 2- and 3- axle trucks of the site. An example is:

```
0.560,33.4,3.7,0.440,59.4,7.4,0.000,0.0,0.0
0.440,40.6,7.4,0.560,66.6,3.7,0.000,0.0,0.0
0.000,0.0,0.0,0.000,0.0,0.0,0.000,0.0,0.0
0.066,20.4,1.5,0.769,34.6,6.8,0.558,30.5,5.9
0.522,26.0,4.9,0.227,39.2,2.2,0.442,37.7,3.5
0.412,38.7,8.6,0.004,54.4,3.7,0.000,0.0,0.0
```

This data is explained as follows:

Class	Row	Weight Axle 1			Weight Axle 2			Weight Axle 3		
		ρ	μ	σ	ρ	μ	σ	ρ	μ	σ
2-Axle	1	0.56	33.4	3.7	0.44	59.4	7.4	0	0	0
	2	0.44	40.6	7.4	0.56	66.6	3.7	0	0	0
	3	0	0	0	0	0	0	0	0	0
	4									
3-Axle	5	0.066	20.4	1.5	0.769	34.6	6.8	0.558	30.5	5.9
	6	0.522	26	4.9	0.227	39.2	2.2	0.442	37.7	3.5
	7	0.412	38.7	8.6	0.004	54.4	3.7	0	0	0

Aw4&5.csv

This file contains the axle weight information for the 4- and 5-axle trucks. It has been found that the axle weights of the 4- and 5-axle trucks depend on the Gross Vehicle Weight (GVW). Thus the data governing these axle weights have been assembled for 12 classes of truck GVW, beginning at 25 kN and increasing in steps of 50 kN.

```
0.0,0.0,0.0,0.0,0.0,0.0
20.9,39.8,39.3,5.2,6.9,7.3
25.6,36.5,38.0,5.4,4.8,5.7
23.9,35.5,40.7,4.3,4.6,5.2
20.3,36.1,43.6,3.6,4.6,5.4
17.4,34.9,47.7,3.0,4.1,5.5
14.8,33.4,51.8,2.1,3.1,4.1
14.5,33.6,51.9,1.5,2.6,3.2
13.9,32.4,53.7,1.3,2.3,3.1
11.9,31.4,56.7,0.9,1.4,0.9
0.0,0.0,0.0,0.0,0.0,0.0
0.0,0.0,0.0,0.0,0.0,0.0
```

```

0.0,0.0,0.0,0.0,0.0,0.0
0.0,0.0,0.0,0.0,0.0,0.0
19.1,36.5,44.5,6.0,7.4,7.2
23.6,32.8,43.7,4.6,4.2,5.0
21.4,33.4,45.3,3.2,4.8,5.4
18.1,33.8,48.1,2.4,4.5,5.5
15.7,32.3,52.0,1.8,3.8,4.7
14.3,31.0,54.6,1.5,3.3,3.9
13.4,29.6,57.1,1.2,2.9,3.4
12.7,27.7,59.6,1.0,2.7,3.1
0.0,0.0,0.0,0.0,0.0,0.0
0.0,0.0,0.0,0.0,0.0,0.0

```

A single line separates the 4- and 5-axle data. The six entries for each line, or GVW range of truck, represent the parameters of the single-mode Normal distributions for the first (W1) and second (W2) axles and the total weight of the tandem or tridem (WT) in the following order:

Mean W1	Mean W2	Mean WT	SD W1	SD W2	SD WT
---------	---------	---------	-------	-------	-------

This has resulted from previous research which has found that the weights of the axles in the tandem or tridem of 4- and 5-axle trucks (respectively) are equal and thus the tandem/tridem may be considered as one weight. The calculated tandem/tridem weight are divided by the number of axles to give each axle a weight in the processing of this data. The values must be separated by commas.

Gross Vehicle Weight

GVWpdf.csv

This file holds the parameters of the distributions that characterize the GVW and speed of each class of truck for both directions. An example of this file is:

```

1,194.5,27.4,0.152,44.2,6.5,0.069,51.2,9.7,0.583,231.1,61.9,0.274,199.9,36.7
0,0,0.395,76.4,20.7,0.887,166.3,53.2,0.24,176.6,29.6,0.553,308.7,49.9
0,0,0.453,117.4,30.5,0.044,268.4,34.7,0.177,331,30.1,0.173,383.2,35.4
1,181.1,22.4,0.143,46.5,8,0.093,56.4,12.4,0.493,243.6,64.6,0.16,205.3,40.1
0,0,0.524,82.9,23.8,0.653,141.5,31.1,0.301,162.1,28.8,0.441,300.6,53.6
0,0,0.333,132.3,31.8,0.254,218.5,33.4,0.206,361.9,31.6,0.399,400.4,35.9

```

Again this is best explained by reference to the following table:

	Speed	2-Axle GVW	3-Axle GVW	4-Axle GVW	5-Axle GVW
Direction 1	3×3	3×3	3×3	3×3	3×3
		Blank Line			
Direction 2	3×3	3×3	3×3	3×3	3×3

In the above table the entry 3×3 refers to the allowance for multi-modal distributions (up to a maximum of three modes) and includes, for each mode, the weight, mean and standard deviation, as explained previously. The values must be separated by commas.

Headway

NHM.csv

Of the headway models, only the HeDS model requires an input file. This model is defined in OBrien & Caprani (2005). An example is:

```
15,0,0,0
0,0.011855673,-0.014268241,0.004048786
0,0.039251526,-0.05978246,0.02212043
70,-0.004412997,0.054824101,-0.066907905
80,-0.004685721,0.052127816,-0.053475193
90,0.001537014,0.020896587,-0.013787689
100,-0.003853623,0.064555837,-0.069172155
110,-0.002530238,0.054511802,-0.059714977
120,-0.001307981,0.048010242,-0.051645258
130,-0.000487752,0.049738587,-0.057875119
140,-0.004995115,0.081041256,-0.086465967
150,-0.004547469,0.080310658,-0.083351351
160,-0.004938412,0.092219287,-0.105416601
170,-0.005000644,0.086893379,-0.097048852
180,0.001987438,0.052114614,-0.058245039
190,0.003366332,0.044909211,-0.063187142
210,0.000379907,0.068461437,-0.077769612
230,-0.006466786,0.117770005,-0.141174818
```

Line 1 indicates the number of flow-dependent headway models (always less than, or equal to, 24). Lines 2 and 3 give the parameters of the quadratic-fit headway cdf for under 1.0 s and between 1.0 s and 1.5 s respectively. The following lines (of number 15 in this example, from Line 1), return the parameters of the quadratic fit to the headway cdf for that flow (trucks per hour) of the first column. The values must be separated by commas.

Constant Traffic Model (Model 1)

This is a simple test model in which the vehicles being generated are constant (i.e. no randomization). It is useful for verification of calculations and vehicle arrivals for example. Each car generated has 2 axles of wheelbase 4.0 m, and weights 1 tonne each. And each truck is a 46 tonne 6-axle semi-trailer, with axle weights 7, 6, 6, and 3×9 tonnes, and spacings 3.5, 2.0, 6.0, 2×1.2 m. The proportion of cars to trucks is as per the traffic flow generation model.

Vehicle Garage Model (Model 2)

This method of generating an artificial traffic stream, is a smoothed bootstrap using variable-bandwidth kernel density estimators approach (Enright, 2010, Scott, 2015). In this method, a vehicle is randomly selected from the garage of measured vehicles (bootstrapping), and then randomised (smoothing) using deviates (kernels). Thus, a very large number of artificial heavy vehicles in the traffic stream, including overloaded vehicles, can be generated, while maintaining intrinsic correlations and relationships within the vehicle database and characteristics (e.g. correlation between axle weights and gross vehicle mass).

The vehicle parameters modified using the kernels are the GVM, axle weights, and axle spacings. The literature presents two usual kernels: a gaussian and triangle kernel. Currently only a gaussian kernel is implemented with indicative parameters for the different vehicle characteristics as shown in the table below.

Vehicle Characteristic	Bias	Standard Deviation (Bandwidth)
Gross vehicle mass	1	0.08
Axle group mass	1	0.05
Axle group spacing	1	0.02

Vehicle Garage File This file is a vehicle file in one of the vehicle file formats (see Appendix). For good simulations it should contain sufficient numbers of each vehicle type to represent the entirety of traffic at the site. The proportions of each vehicle type should also reflect the relative proportions of vehicle types at the site.

Kernel File The kernel file is a csv type formatted file, corresponding to the table above as follows:

```
1.0,0.08
1.0,0.05
1.0,0.02
```

3.3 Configuration File

The user interacts with the program through the configuration file.

Important!

The input file must be called “BTLSin.txt” and it must be in the working folder.

An example input file is shown next, and each input line explained following.

Line	BTLSin.txt
Line	BTLSin.txt
1	// -----
2	// START OF BRIDGE TRAFFIC LOAD SIMULATION INPUT
3	// -----
4	//
5	// -----
6	// *** INPUT SPECIFICATIONS ***
7	// -----
8	//
9	// Program Mode (1 - Gen & Sim, 2 - Gen, 3 - Read & Sim)
10	1
11	//
12	// TRAFFIC GENERATION PARAMETERS
13	// -----
14	// No. of days of traffic simulation:
15	20
16	// Location of Traffic folders
17	C:\Users\ccapr\Google Drive\~Research\Workings\Traffic\Auxerre
18	// Default truck track width (cm)
19	190.0
20	// Standard deviation of eccentricity in lane (cm) (about 20 cm usually)
21	0
22	// Vehicle Generation Model to be used
23	// (0 - Grave, 1 - Nominal, 2 - Garage)
24	2
25	// Headway model to be used:
26	// (0 - Auxerre HeDS (truck only), 1 - Constant, 5 - Congestion, 6 - free-flow)
27	6
28	// Traffic Classification System to be used:
29	// (0 - No. of Axles, 1 - Axle Pattern)
30	1
31	// Lane and flow definition file:
32	LaneFlowData_80.csv
33	// Nominal congested spacing, front to back (m):
34	5
35	// Congested speed (km/h):
36	30
37	// Congested gaps coefficient of variation:
38	0.05
39	//
40	// TRAFFIC GARAGE PARAMETERS
41	// -----
42	// Garage file as basis for generation (in "Traffic input file format" below):
43	garage.txt
44	// Kernels (mean, std) file for randomizing the garage vehicles (lines in order GVW, AW, AS)
45	kernels.csv
46	//
47	// TRAFFIC INPUT FILE PARAMETERS
48	// -----
49	// Traffic input file to be analysed:
50	WGBSlowLanes_Jul.txt
51	// Traffic input file format (CASTOR - 1, BeDIT - 2, DITIS - 3, MON - 4, SiWIM - 5):
52	4
53	// Impose constant speed on all vehicles (1 or 0):
54	0
55	// Use average speed of vehicles in file if constant speed imposed (1 or 0)
56	1
	// Constant speed of vehicles if not average used (km/h):

\\ Comments: The program reads all lines of the configuration file except those preceded with C++ style commenting: “\\”. The user is free to add further commenting to the file as they wish, once the order of the input variables is not altered.

Important!

Depending on the program mode, some inputs are redundant. However, they must still be specified as ‘placeholders’ to keep the order of inputs the same.

Line 1: The user specifies the program mode using 1, 2, or 3 for the modes as defined on page 15.

Line 2: Specify the number of days of traffic to simulate in Modes 1 or 2. This input is redundant in the case of Mode 3 when the traffic file is specified.

Line 3: Specify the location of the traffic generation files used in Program Modes 1 and 2 for the Grave Traffic Model (Model 0).

Line 4: Specify the truck track width (axle width) to be used if there is no ATW.csv file for the site in the folder of Line 3 – only relevant for influence surface calculations.

Line 5: Specify the standard deviation of the eccentricity of the vehicles in their lane. This approach assumes a mean of zero and a normal distribution. Some site data suggests about 20 cm is a reasonable value. Again, only applies to influence surface calculations.

Line 6: Vehicle generation model to be used:

- Model 0: The Grave Model, as described in Caprani (2005);
- Model 1: A constant vehicle model, one car and one truck type;
- Model 2: Bootstrap from vehicle garage using kernels.

Line 7: Specify the headway model to be used in the generation of artificial traffic. The options are (note the odd-numbering for ‘historical’ reasons):

- “0” – The HeDS (Headway Distribution Statistics) model of OBrien & Caprani (2005). This is suitable for the Auxerre site-measured flowrates only. It is a free-flow model that generates only trucks.
- “1” – The Constant headway model.
- “5” – Congestion model as per Caprani (2012), summarized in the following diagram. A nominal axle gap is specified (Line 6), along with a coefficient of variation between successive vehicles in all lanes (i.e. trucks and cars) (Line 8) and gaps are then generated using a normal distribution.



- “6” – Free-flow model which uses a Poisson arrival assumption based upon the Normalized Headway Model of Crespo-Minguillón and Casas (1997). This accounts for different flow rates (Q) as follows:

$$F(t) = 1 - e^{-\lambda t}$$

Where $1/\lambda$ is the mean headway, i.e. the average time gap between vehicles in the hour of the current total (cars and trucks) flowrate Q and so is given by $Q/3600$.

For all models the program checks that no overlapping of vehicles can occur by ensuring the generated gap is greater than the required minimum gap (taking account of the maximum bridge length, vehicle lengths, and speed difference between them).

Line 8: The traffic classification scheme to be used in output:

- 0 – classify by the number of axles
- 1 – classify by the axle pattern, where axle groups are identified as consecutive axles with spacing < 2.1 m.

Line 9: Specify the name of the Lane Flow definition file that is in the working folder.

Line 10: Specify the nominal axle gap for Headway Model 5 – congestion.

Line 11: Specify the speed of all vehicles for Headway Model 5 – congestion. Note that this, combined with the calculation time step (Line 16), effectively renders a distance-stepping algorithm and so this speed can be notional to achieve a required distance step.

Line 12: Specify the coefficient of variation of the nominal congested gap for Headway Model 5 – congestion.

Line 13: Specify the filename in the working folder for the garage of vehicles to be used for Traffic Generation Model 2. Note, this file must be in the format of the Traffic Input File specified in Line 16.

Line 14: Specify the filename in the working folder of the kernel specifications for the Traffic Generation Model 2.

Line 15: Specify the name of the traffic input file in the working folder for Program Mode 3.

Line 16: Specify the format of the input file: 1 – CASTOR format, 2 – BeDIT, 3 – DITIS format, 4 – MON, 5 – SiWIM CSV (introduced in v1.3.7, input only). See Appendix for traffic file format definitions.

Line 17: BTLS normally passes each vehicle across the bridge according to its own speed when in Program Mode 3. Setting this option to “1” imposes constant speed on all vehicles for comparison with some other algorithms – mostly to do with congestion traffic files.

Line 18: When constant speed is imposed (Line 17), if this option is “1” then the average speed of all vehicles in the file will be used, otherwise the speed specified in Line 19 will be used.

Line 19: Specifies the constant speed if Line 17 is “1” and Line 18 is “0”.

Line 20: Specify the name of the bridge definition file (see Section 0) in the working folder.

Line 21: Specify the name of the influence line definition file (see Section 3.6) in the working folder.

Line 22: Specify the name of the influence surface definition file (see Section 3.7) in the working folder.

Line 23: Specify the calculation time step which is used in passing the vehicles over the bridges. 0.1 s has been found a good compromise between accuracy and efficiency. For some very sharp influence lines (e.g. shear forces) a finer step may be required. A sensitivity study is recommended.

Line 24: To avoid unnecessary computation of smaller vehicles, this specifies the minim GVW for a vehicle's load effect to be calculated. Its spatial arrangement on the road is not affected if its GVW is less than this number. The units are deci-tonnes (t/10)

Line 25: Specify "1" to write a full time history of the load effects – see section on BTLS Output for more details. This should be set to "0" for long simulations due to enormous resulting file size and slow execution.

Line 26: Specify "1" to write the load effect value for each loading event that occurs. Again this can be a large file and cause slow execution for long-run simulations. See section on BTLS Output for more details.

Line 27: If each loading event value is to be written (Line 24), this option if set to "1" specifies the number of events that are stored in memory before writing to the hard drive. See section on BTLS Output for more details (Section 5).

Line 28: Writes a file suitable for further fatigue calculations, giving load cycles.

Line 29: Conduct Rainflow analysis on the load effects, which is for fatigue research.

Line 30: The number of decimals kept for Rainflow analysis, acting as the bin size.

Line 31: The cutoff value for Rainflow analysis.

Line 32: Rainflow-output file buffer size.

Line 33: For all program modes, this option if "1" will write the generated or read-in vehicle file. For long run simulations this should be "0" as very large files can result, filling hard drive space and causing very slow computation. Mostly useful for short debugging or test runs.

Line 34: This specifies the format of the output file traffic file: 1 – CASTOR format, 2 – BeDIT, 3 – DITIS, 4 – MON. Note that the SiWIM CSV format (5) is read-only and cannot be used for output.

Line 35: The name of the file to be written if Line 33 is "1".

Line 36: If the vehicle file is to be written (Line 33 set to "1"), this specifies the number of vehicles that are stored in memory before writing to the hard drive. See section on BTLS Output for more details.

Line 37: If this is set to "1" files are output giving the traffic flow and composition information for each hour of the simulation for each lane.

Line 38: If this option is "1" calculations are performed that can be used to write block maxima output. Set to "0" to override all block maxima output and calculations.

Line 39: For block maximum output, this specifies the block size in days for which the maximum is retained (it can be zero if Line 40 has a number > 0).

Line 40: For block maximum output, this specifies the block size in seconds for which the maximum is retained (it can be zero if Line 39 has a number > 0).

Line 41: Specify “1” to write block maximum load effect and vehicle output files for each number of vehicles comprising the events. See section on BTLS Output (Section 5.4) for more details.

Line 42: Specify whether to write the block maximum summary files (“1” or “0”). See section on BTLS Output (Section 5.4) for more details.

Line 43: Specify “1” to write block maximum load effect output files for which the events are not separated by the number of vehicles in the event, or “0” to not. See section on BTLS Output (Section 5.4) for more details.

Line 44: If block maximum output is to be written (Line 38), this option if set to “1” specifies the number of events that are stored in memory before writing to the hard drive. See section on BTLS Output (Section 5.4) for more details.

Line 45: If this option is “1” calculations are performed that can be used to write peaks-over-threshold (POT) output. Set to “0” to override all POT output and calculations. Note that the thresholds for each load effect are set in the Bridge Definition File (Section 0).

Line 46: For POT output, this specifies if the vehicles comprising the peak events are to be output to a vehicle-event file. See section on BTLS Output (Section 5.5) for more details.

Line 47: For POT output, this specifies if summary files are to be written. See section on BTLS Output (Section 5.5) for more details.

Line 48: For POT output, this specifies if a peaks counter file is to be written. See section on BTLS Output (Section 5.5) for more details.

Line 49: For peaks counter output, this specifies the interval size in days for which the peaks are counted (it can be zero if Line 50 has a number > 0).

Line 50: For peaks counter output, this specifies the interval size in seconds for which the peaks are counted (it can be zero if Line 49 has a number > 0).

Line 51: If POT output is to be written (Line 41), this option if set to “1” specifies the number of events that are stored in memory before writing to the hard drive. See section on BTLS Output (Section 5.5) for more details.

Line 52: If this option is “1” calculations are performed that accumulate simple statistics of load effect and vehicles throughout the simulation. Set to “0” to override all statistics output and calculations.

Line 53: This specifies if the statistics for each load effect accumulated through the whole simulation are to be output. See section on BTLS Output (Section 5.6) for more details.

Line 54: This specifies if the statistics for each load effect are to be output at particular time intervals. See section on BTLS Output (Section 5.6) for more details.

Line 55: If interval statistics are to be output, this specifies the interval duration. See section on BTLS Output (Section 5.6) for more details.

Line 56: If interval statistics are to be output, this specifies the number of intervals that are stored in memory before writing to file.

3.4 Bridge Definition File

The bridges over which vehicles are to pass are defined in the bridge definition file, specified in the configuration file (BTLSin.txt). The file name is arbitrary.

Important!

A bridge definition file must be included in the working folder.

An example bridge definition file for a single bridge with 4 load effects is shown:

Line No.	File
1	1, 40.0, 2, 4
2	1, 1, 3000.0
3	1, 1, 0.728125, 0.271875
4	2, 1, 3000.0
5	2, 1, 0.728125, 0.271875
6	3, 2, 3000.0
7	2, 2, 0.728125
8	2, 2, 0.271875
9	4, 3, 3000.0
10	1

Any number of bridges can be defined, as can any number of load effects for each bridge. Each bridge definition is formatted as follows:

Bridge Information: (e.g. “1, 40.0, 2, 4”) This first line specifies some general information about the bridge:

- Column 1: the bridge number, a positive integer – 1 in this case;
- Column 2: the span of the bridge in metres, a real positive number – 40.0 m in this case;
- Column 3: the number of lanes on the bridge, a positive integer – 2 in this case;
- Column 4: the number of load effects to be considered for this bridge, a positive integer – 4 in this case.

Each load effect is then defined on the following lines with a format depending on the type of load effect. The first line in each case is the basic information, formatted as follows:

Load Effect Information: (e.g. “1, 1, 3000.0”)

- Field 1: the load effect number, a positive integer – e.g. 1.
- Field 2: the load effect definition type:
 - 1 – lane factors are applied to a single influence line;
 - 2 – separate influence lines and lane weights are applied to each lane;
 - 3 – an influence surface is used for the load effect.
- Field 3: the threshold to be used for this load effect in peaks-over-threshold analysis, e.g. 3000.0 kNm. If this number is absent it is assumed to be zero.

The line(s) following the load effect information line are formatted differently depending on the load effect definition type, 1, 2, or 3, as follows:

Load Effect Definition Type 1 – Lane factors applied to single IL

Only one line is required in this case, for example “1, 1, 0.728125, 0.271875” (line 3) in the file above. This line is formatted as follows:

- Field 1: The type of influence line:
 - 1 – if it is a built-in influence line;
 - 2 – if it is a user-defined discrete influence line.
- Field 2: The influence line number (whether built-in or discrete).
- Fields3+: the lane factors to be applied to the influence line for Lane 1, 2 etc.

Line 5 in the file above shows that a discrete influence line is being used for Load Effect 2 with lane factors for all lanes.

Load Effect Definition Type 2 – Separate ILs and lane weights for each lane

In this case a line is required, corresponding to each lane, formatted as follows:

- Field 1: The type of influence line:
 - 1 – if it is a built-in influence line;
 - 2 – if it is a user-defined discrete influence line.
- Field 2: The influence line number (whether built-in or discrete).
- Fields3: the lane factor to be applied to the influence line for this lane.

In the example file above, load effect 3, starting on line 6, is using this form of definition. Line 7 defines that Lane 1 uses discrete IL number 2 with lane weight 0.728125. Similarly, line 8 defines that Lane 2 uses the same discrete IL number 2 with lane weight 0.271875.

Load Effect Definition Type 3 – An influence surface is used for the bridge

In this case only a single line is required, assigning an influence surface number to the load effect. Thus, in the above file, line 9 shows that load effect 4 is using an influence surface, whilst line 10 assigns influence surface 1 to it.

Lane factors represent the proportion of load of the corresponding lane (i.e. lane factor 3 is for lane 3) that contributes to the load effect in the element under consideration. In this way, an influence surface is effectively defined as slices along each lane. Note that this model means that the influence surface must be a scaled version of itself transversely across the bridge – this is not always the case however.

Finally, it can be noted that the above example file is a test file: all of the load effects use different means of arriving at the same outcome once the discrete influence line is for mid-span bending moment, and the influence surface is that given as the example influence surface later on.

Built-In Influence Functions

The built-in influence functions are mathematical expressions that apply for any bridge length and can be weighted with any value of lane factor. Consequently, these built-in functions execute more quickly than read-in influence lines.

The description and index for the built in functions are:

Index	Influence Line	Location
1	Mid-span bending moment for a simply supported beam	B
2	Bending moment over the central support of a two-span beam	E
3	Left-hand shear in a simply-supported beam	A
4	Right-hand shear in a simply-supported beam	C
5	Left-hand shear for a two-span beam	D
6	Right-hand shear for a two-span beam	F
7	Total amount of load on the bridge (i.e. the unit influence line)	
8	Bending moment at the 2 nd support of a three-span beam	G
9	Bending moment at the 3 rd support of a three-span beam	H

A

B

C

D

E

F

G

000000000

H

3.5 Lane Flow Data File

This file contains all information relating to the number of lanes, the flow in each lane throughout the day, and the traffic composition. It applies when artificial traffic is being created using one of the free-flow headway models.

Important!

A lane flow definition file must be included in the working folder.

This file must be in *.csv format (but can have a *.txt extension). In *.csv format it is easily edited in a spread sheet program as shown:

	A	B	C	D	E	F	G	H	I
1	1	1							
2	0	153.8	248	10	80	23	2.8	31.7	42.5
3	1	131	248	10	80	23	2.8	31.7	42.5
4	2	131.8	248	10	80	23	2.8	31.7	42.5
5	3	123.8	248	10	80	23	2.8	31.7	42.5
6	4	114	248	10	80	23	2.8	31.7	42.5
7	5	121.2	248	10	80	23	2.8	31.7	42.5
8	6	141.2	248	10	80	23	2.8	31.7	42.5
9	7	155.4	248	10	80	23	2.8	31.7	42.5
10	8	154	248	10	80	23	2.8	31.7	42.5
11	9	141	248	10	80	23	2.8	31.7	42.5

Each lane of the simulation is defined by 25 rows of data:

- The first row defines the lane number (sequential) and direction number (1 or 2) in columns 1 and 2 (or A and B in the screenshot);
- The next 24 rows describe the traffic flow for each hour of the day (i.e. rows 2, 3, 4... in the screenshot above).

Note that it is assumed that every day of the simulation is the same. Typically only economic days of traffic are simulated of 5 days per week, 50 weeks per year (250 days per year). It is assumed that each such day has the same properties.

The structure of each hour description is as follows, with numbers given from hour 0 of the screen shot above (at row 2):

- Column 1: The hour identifier, starting at midnight, 0 to 23 (e.g. 0)
- Column 2: The mean truck flow rate in this hour (trucks/hour) (e.g. 153.8)
- Column 3: Mean velocity of traffic (dm/s) (e.g. 248)
- Column 4: Standard deviation of the velocity (dm/s) (e.g. 10)
- Column 5: Percentage of cars in this traffic model (e.g. 80)
- Column 6: Percentage of trucks that are 2 axle (e.g. 23)
- Column 7: Percentage of trucks that are 3 axles (e.g. 2.8)
- Column 8: Percentage of trucks that are 4 axles (e.g. 31.7)
- Column 9: Percentage of trucks that are 5 axles (e.g. 42.5)

The rationale for having the input in this form is the ease of altering the percentage cars and overall flow rate without modifying the truck flow and composition. This is best explained through an example of the calculations the program performs:

1. From Column 5, the truck percentage is $100 - 80 = 20\%$;
2. This 20% represents a flow of 153.8 vehicles per hour (Column2);
3. Thus the total flow rate is $153.8 / 0.2 = 769$ vehicles per hour.
4. Of the 20% vehicles that are trucks, for example, 42.5% are 5-axle trucks, thus there will be $0.425 * 153.8 = 65.4$ 5-axle trucks on average for this hour.

Changing Column 4 then changes the overall flow rate, without changing the number of trucks that arrive.

Note that a normal distribution is assumed for the speed of all vehicles.

Each lane to be included in the simulation must have the above information. An example file with two lanes, one in each direction is given below:

```
1,1,, , , ,
0,153.8,248,10,80,23,2.8,31.7,42.5
1,131,248,10,80,23,2.8,31.7,42.5
2,131.8,248,10,80,23,2.8,31.7,42.5
3,123.8,248,10,80,23,2.8,31.7,42.5
4,114,248,10,80,23,2.8,31.7,42.5
5,121.2,248,10,80,23,2.8,31.7,42.5
6,141.2,248,10,80,23,2.8,31.7,42.5
```

7,155.4,248,10,80,23,2.8,31.7,42.5
8,154,248,10,80,23,2.8,31.7,42.5
9,141,248,10,80,23,2.8,31.7,42.5
10,126.4,248,10,80,23,2.8,31.7,42.5
11,101.6,248,10,80,23,2.8,31.7,42.5
12,95.8,248,10,80,23,2.8,31.7,42.5
13,88.2,248,10,80,23,2.8,31.7,42.5
14,93,248,10,80,23,2.8,31.7,42.5
15,109,248,10,80,23,2.8,31.7,42.5
16,124.2,248,10,80,23,2.8,31.7,42.5
17,151,248,10,80,23,2.8,31.7,42.5
18,141.8,248,10,80,23,2.8,31.7,42.5
19,172.2,248,10,80,23,2.8,31.7,42.5
20,141.4,248,10,80,23,2.8,31.7,42.5
21,148,248,10,80,23,2.8,31.7,42.5
22,157.2,248,10,80,23,2.8,31.7,42.5
23,159.4,248,10,80,23,2.8,31.7,42.5
2,2,,,,
0,92.2,222,10,80,21.9,2.3,31,44.8
1,79.6,222,10,80,21.9,2.3,31,44.8
2,67,222,10,80,21.9,2.3,31,44.8
3,74.8,222,10,80,21.9,2.3,31,44.8
4,81.6,222,10,80,21.9,2.3,31,44.8
5,94.8,222,10,80,21.9,2.3,31,44.8
6,102.4,222,10,80,21.9,2.3,31,44.8
7,121.2,222,10,80,21.9,2.3,31,44.8
8,127.4,222,10,80,21.9,2.3,31,44.8
9,127.2,222,10,80,21.9,2.3,31,44.8
10,112.2,222,10,80,21.9,2.3,31,44.8
11,111.4,222,10,80,21.9,2.3,31,44.8
12,110.2,222,10,80,21.9,2.3,31,44.8
13,146,222,10,80,21.9,2.3,31,44.8
14,160.4,222,10,80,21.9,2.3,31,44.8
15,152,222,10,80,21.9,2.3,31,44.8
16,151,222,10,80,21.9,2.3,31,44.8
17,167.2,222,10,80,21.9,2.3,31,44.8

18,179.6,222,10,80,21.9,2.3,31,44.8
 19,164.8,222,10,80,21.9,2.3,31,44.8
 20,206.6,222,10,80,21.9,2.3,31,44.8
 21,228.4,222,10,80,21.9,2.3,31,44.8
 22,189,222,10,80,21.9,2.3,31,44.8
 23,138.8,222,10,80,21.9,2.3,31,44.8

3.6 Influence Line Definition File

This file stores the definitions of any discrete influence lines that are required. It must be in *.csv format (but can have a *.txt extension).

Important!

An influence line definition file must be included in the working folder if it is required for the analysis.

The first line of the file is the number of influence lines defined within. Subsequently, each influence line is defined with the following structure:

- A first line giving the influence line number (Column 1) and the number of points defining the influence line (Column 2);
- Subsequent lines define the influence line using x , y , pairs for the location and ordinate values (Columns 1 and 2 respectively).

Discrete influence line processing takes longer than built-in expressions. The program must search the vector of x -coordinates to find the points surrounding the axle location. Linear interpolation of the ordinates is then used to find the ordinate at the axle location. The spacing of points need not be uniform. Therefore, prefer to use as few points as is necessary where the influence line is linear, and more points where it is curved.

Important!

The program warns if the last x -coordinate is not the same as the length of the bridge defined in the Bridge Definition file. Behaviour in this case is generally unpredictable. However this warning can be issued due solely to rounding, and in this case no problems have been observed.

An example file is given below:

- the first influence line is a test of a 40 m simply-supported mid-span bending moment calculation,
- the second is an influence line from the Millau viaduct, courtesy if IFSTTAR, France.

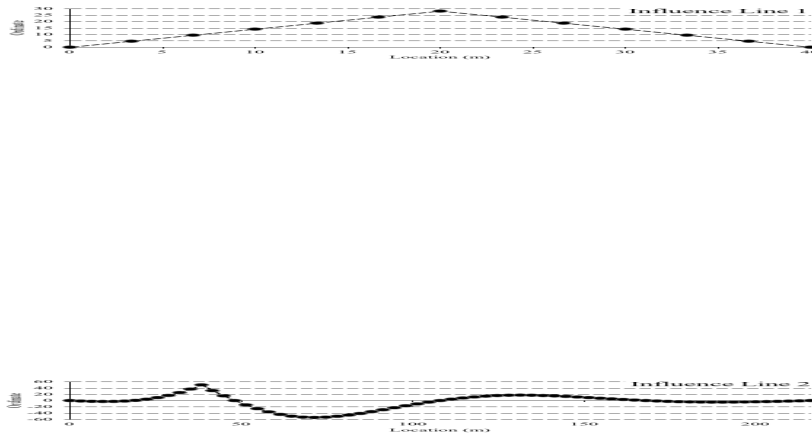
```
2
1, 13
0.000000, 0.000000
3.333333, 4.695709
6.666667, 9.391137
10.000000, 14.086847
13.333333, 18.782556
16.666667, 23.477984
20.000000, 28.173693
```

23.333333, 23.477984
26.666667, 18.782556
30.000000, 14.086847
33.333333, 9.391137
36.666667, 4.695709
40.000000, 0.000000
2, 67
0.000000, 0.000000
3.200000, -1.404704
6.400000, -2.482683
9.600000, -2.967398
12.800000, -2.765344
16.000000, -1.658346
19.200000, 0.578218
22.400000, 4.170048
25.600000, 9.341767
28.800000, 16.319074
32.000000, 25.326593
35.200000, 36.574975
38.400000, 50.127631
41.600000, 31.722458
44.800000, 15.056238
48.000000, 0.000000
51.333333, -13.793401
54.666667, -25.918783
58.000000, -36.084887
61.333333, -44.196060
64.666667, -49.871840
68.000000, -52.900498
71.333333, -53.815114
74.666667, -52.833863
78.000000, -50.155575
81.333333, -46.221973
84.666667, -41.193196
88.000000, -35.436810
91.333333, -29.230102

94.666667, -22.847134
98.000000, -16.583464
101.333333, -10.646519
104.666667, -5.083589
108.000000, 0.000000
111.333333, 4.397896
114.666667, 8.216499
118.000000, 11.460109
121.333333, 14.100781
124.666667, 15.951508
128.000000, 16.941357
131.333333, 17.243363
134.666667, 16.930610
138.000000, 16.067582
141.333333, 14.799371
144.666667, 13.179716
148.000000, 11.327914
151.333333, 9.335319
154.666667, 7.293285
158.000000, 5.293167
161.333333, 3.385477
164.666667, 1.617506
168.000000, 0.000000
171.200000, -1.360639
174.400000, -2.545019
177.600000, -3.539167
180.800000, -4.268925
184.000000, -4.746116
187.200000, -4.978264
190.400000, -4.989011
193.600000, -4.803078
196.800000, -4.444110
200.000000, -3.936826
203.200000, -3.307020
206.400000, -2.577262
209.600000, -1.773345

212.800000, -0.904943

216.000000, 0.000000



3.7 Influence Surface Definition File

This file stores the definitions of any influence surfaces that are required. It must be in *.csv format (but can have a *.txt extension).

Important!

An influence surface definition file must be included in the working folder if it is required for the analysis.

Any number of influence surfaces can be defined in the file. Each influence surface is defined on a rectangular grid according to a specific format as follows:

- The first line defines the basic data as follows:
 - Field 1: the influence surface number;
 - Field 2: the number of rows of data (the number of x - or longitudinal coordinates) specifying the influence surface grid points;
 - Field 3: the number of lanes on the bridge/influence line
 - Field 3+ i : the location of the edge of lane i on the influence surface;
 - Last Field: the upper edge of the last lane.
- Line 2 defines the y - or transverse coordinates of the grid points.
- Each row then defines the ordinates at the grid points, the first value of which is the x -coordinate.

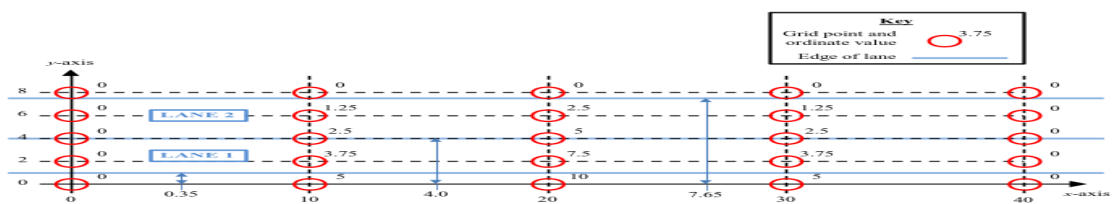
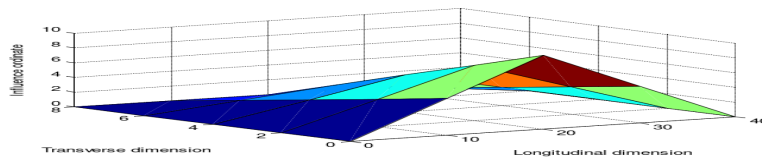
It is taken that Direction 1 traffic travels in the positive x -direction. For load application between grid points, linear interpolation is used in both directions. Hence, for flat areas of the influence surface a larger grid size can be used, but for more peaked zones, a more refined grid is required.

An example file is shown below in which a single test influence surface for mid-span bending moment of an edge beam is defined for comparison with a standard method using built-in or discrete influence lines and lane factors. The following figure describes the file and its definitions.

```

1,5,2,0.35,4.0,7.65
0,0,2,4,6,8
0,0,0,0,0,0
10,5,3.75,2.5,1.25,0
20,10,7.5,5,2.5,0
30,5,3.75,2.5,1.25,0
40,0,0,0,0,0

```



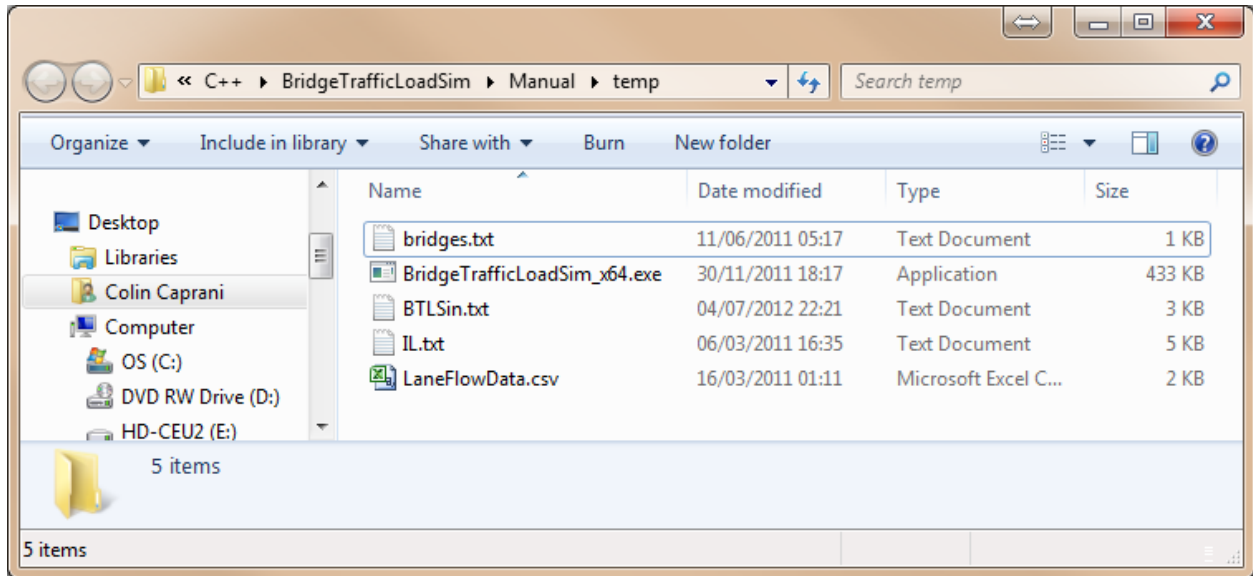
Important!

If multiple influence surfaces are defined for a bridge, the program assumes that the lane coordinates are the same for all influence surfaces. This is as it should be for a single real bridge of course. However, different lane coordinates can be used with different influence surface definitions for different bridges under the same traffic, either read or generated.

4 Using BTLS

4.1 Running the program

The program can be run from any folder as explained previously. An example working folder showing all files necessary to execute the program is given below:



Note that the 64 bit version is being used here.

Important!

To run the program, double click the executable (*.exe) file on Windows, or execute `./BTLS` from a terminal in the working folder on Linux and macOS.

As of v1.3.7, if the configuration file `BTLSin.txt` cannot be found in the working folder, the program prints

```
***ERROR: BTLSin file could not be opened: BTLSin.txt
```

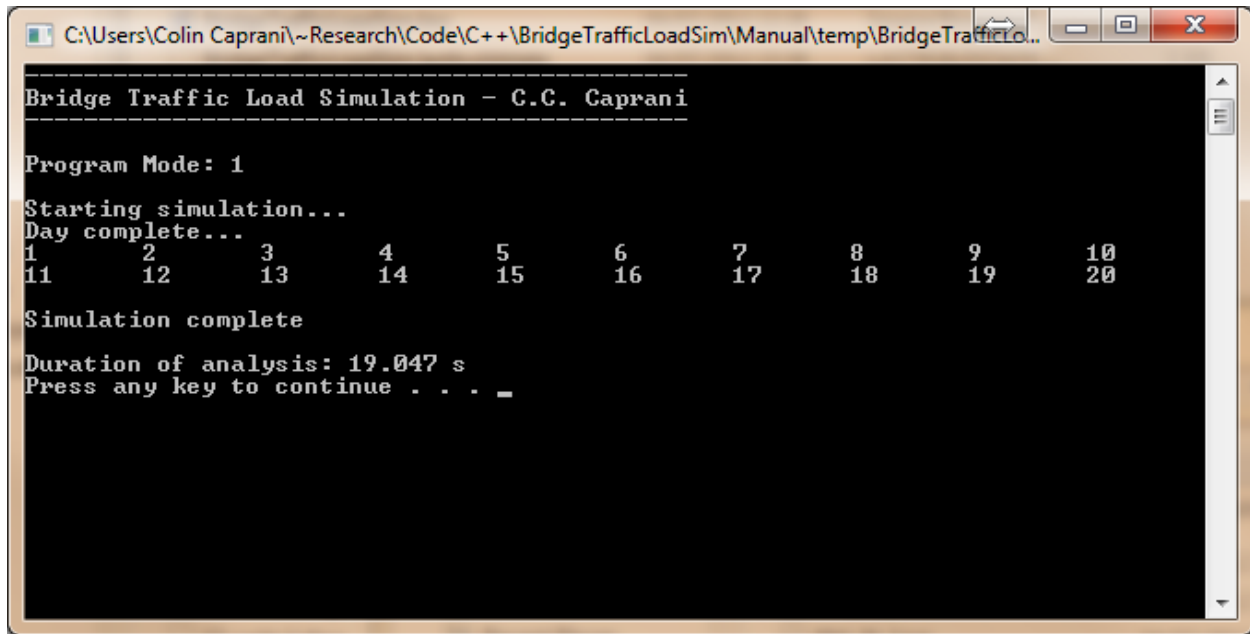
and exits immediately with a non-zero exit code. (Earlier versions attempted to continue with built-in default values, which generally led to a crash.) On success the program returns exit code 0, so BTLS can be scripted reliably in batch runs.

4.2 Console output

Some examples of console output during program execution are given.

Example 1: Program Mode 1

After each day of simulation is complete, the program outputs a notice. At the end of the simulation the elapsed time is displayed for information. In this case, 20 days of traffic and generated and simulated crossing 5 bridges, each with 3 load effects.



```
Bridge Traffic Load Simulation - C.C. Caprani

Program Mode: 1
Starting simulation...
Day complete...
1      2      3      4      5      6      7      8      9      10
11     12     13     14     15     16     17     18     19     20
Simulation complete
Duration of analysis: 19.047 s
Press any key to continue . . . _
```

Example 1: Program Mode 2

Ten days of vehicles are generated and the current day number is given (right hand columns). Each time the program flushed the vehicle buffer to the file, an output is given of the simulation time at which it occurred.

```

C:\Users\Colin Caprani\~Research\Code\C++\BridgeTrafficLoadSim\Manual\temp\BridgeTrafficLo...
Bridge Traffic Load Simulation - C.C. Caprani
-----
Program Mode: 2
Starting simulation...
Day complete...

Flushing buffer of 10000 vehicles at 1/1/0 9:11:36
Flushing buffer of 10000 vehicles at 1/1/0 17:45:52      1
Flushing buffer of 10000 vehicles at 2/1/0 0:19:27
Flushing buffer of 10000 vehicles at 2/1/0 7:33:4
Flushing buffer of 10000 vehicles at 2/1/0 14:45:28
Flushing buffer of 10000 vehicles at 2/1/0 22:2:8        2
Flushing buffer of 10000 vehicles at 3/1/0 5:20:23
Flushing buffer of 10000 vehicles at 3/1/0 12:33:24
Flushing buffer of 10000 vehicles at 3/1/0 19:45:18      3
Flushing buffer of 10000 vehicles at 4/1/0 2:54:54
Flushing buffer of 10000 vehicles at 4/1/0 10:8:59
Flushing buffer of 10000 vehicles at 4/1/0 17:23:3
Flushing buffer of 10000 vehicles at 5/1/0 0:35:47
Flushing buffer of 10000 vehicles at 5/1/0 7:48:19
Flushing buffer of 10000 vehicles at 5/1/0 15:0:21
Flushing buffer of 10000 vehicles at 5/1/0 22:12:27      5
Flushing buffer of 10000 vehicles at 6/1/0 5:22:41
Flushing buffer of 10000 vehicles at 6/1/0 12:41:41
Flushing buffer of 10000 vehicles at 6/1/0 19:52:39      6
Flushing buffer of 10000 vehicles at 7/1/0 3:12:26
Flushing buffer of 10000 vehicles at 7/1/0 10:21:49
Flushing buffer of 10000 vehicles at 7/1/0 17:32:31      7
Flushing buffer of 10000 vehicles at 8/1/0 0:47:2
Flushing buffer of 10000 vehicles at 8/1/0 7:59:48
Flushing buffer of 10000 vehicles at 8/1/0 15:14:25
Flushing buffer of 10000 vehicles at 8/1/0 22:24:9       8
Flushing buffer of 10000 vehicles at 9/1/0 5:36:6
Flushing buffer of 10000 vehicles at 9/1/0 12:48:16
Flushing buffer of 10000 vehicles at 9/1/0 20:5:50       9
Flushing buffer of 10000 vehicles at 10/1/0 3:10:33
Flushing buffer of 10000 vehicles at 10/1/0 10:25:16
Flushing buffer of 10000 vehicles at 10/1/0 17:36:52     10

Flushing buffer of 8770 vehicles at 11/1/0 0:0:2
Simulation complete

Duration of analysis: 4.888 s
Press any key to continue . . . _

```

The folder is then populated with the outputted vehicle file as named in “BTLSin.txt”.

Example 3: Program Mode 3

The traffic file created in the last example is read in and passed over 5 bridges, each with 3 load effects. The output is as follows:

```

C:\Users\Colin Caprani\~\Research\Code\C++\BridgeTrafficLoadSim\Manual\temp\BridgeTrafficLo...
Bridge Traffic Load Simulation - C.C. Caprani
-----
Program Mode: 3
Reading traffic file...
Starting simulation...
Day complete...
1      2      3      4      5      6      7      8      9      10

Simulation complete
Duration of analysis: 32.025 s
Press any key to continue . . . _

```

Note the slower execution time than for Program Mode 1 which has 20 days of traffic. This is caused by the additional overhead required to manipulate the 25 MB traffic file that has been read into memory.

4.3 Input Errors

When there are errors in the input, the behaviour is unpredictable. More informative user feedback will be built in soon.

Note that a missing `BTLSin.txt` is detected and reported cleanly: the program exits with an error message and a non-zero exit code (see Section 4.1).

Some potential problems:

- The supporting files (e.g. bridge, lane flow, IL) cannot be found in the working folder;
- The traffic folder cannot be found (it should be at “C:\Traffic”);
- The vehicle file to be read in cannot be found (Program Mode 3).
- Outputs are not matched to Program Mode (e.g. Program Mode 2 – Generate vehicle file, but no vehicle file is to be output – Line 20 of `BTLSin.txt`.)

In each of these cases, the program may:

- Output helpful warnings;
- Flash open and close immediately;
- Remain open and display unusual text, such as “Conversion Error”.

Admittedly, all behaviours should be of the first type – this will improve.

5 BTLS Output

5.1 Introduction

BTLS can produce large amounts of output, especially for long run simulations or heavily congested traffic. Accessing the hard drive often can significantly slow the program’s execution. Therefore, keep buffer sizes as large as memory and execution speed can allow.

All outputs are in text file format with specific information layouts and formats for each type of output file.

Effect on Execution Speed

Obviously, the more outputs that are needed the slower the simulation. Some general comments to aid execution time are:

- Only output what is needed: this can be ascertained by a few short runs before doing the main long-run simulation.
- Use the buffer size variables to good effect: for sample runs monitor the program's RAM usage, increasing the buffer sizes as much as possible to prevent undue writing to disc, one of the slowest operations.
- The statistics output is intended mainly for short runs to give information for peaks-over-threshold analysis (i.e. threshold levels). Consequently, turn it off when it is not required.
- Similarly, the flow data output is only intended for short-run verification that the traffic is being properly modelled. Consequently, turn it off when it is not required.
- For the Block Max and POT outputs, turn off the vehicle output files if they are not needed.

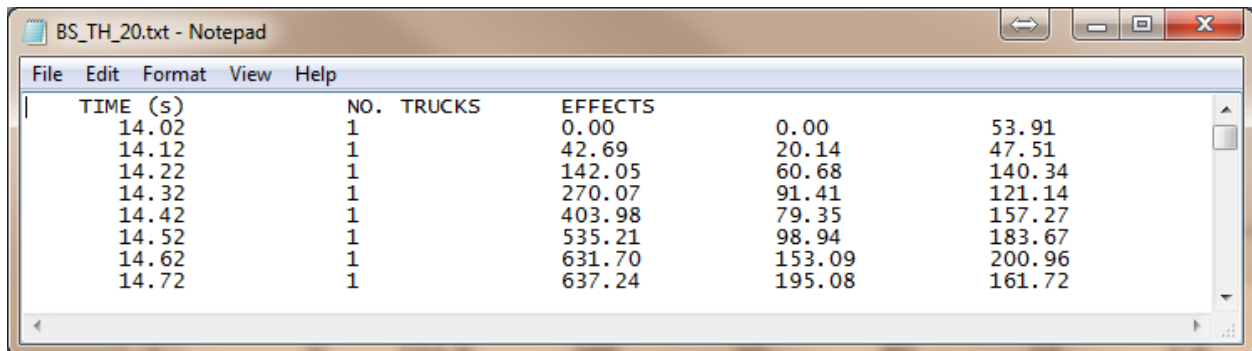
5.2 Miscellaneous Output

Time History File

If a full time history is selected to be output (Line 25 of BTLSin.txt), BTLS creates a single file for each bridge named:

TH_*L*.txt

Where *L* is the bridge length. The file gives the load effect at each time step of the simulation for each load effect considered for the bridge. Note that no output is given when there is zero load effect. A sample output is:



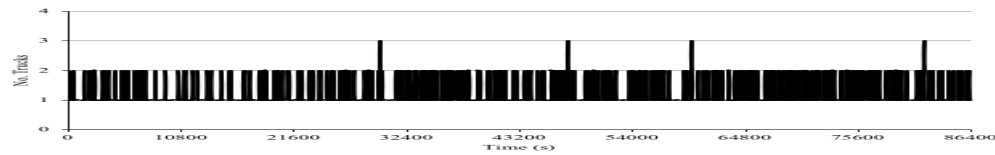
TIME (s)	NO. TRUCKS	EFFECTS				
14.02	1	0.00	0.00	53.91		
14.12	1	42.69	20.14	47.51		
14.22	1	142.05	60.68	140.34		
14.32	1	270.07	91.41	121.14		
14.42	1	403.98	79.35	157.27		
14.52	1	535.21	98.94	183.67		
14.62	1	631.70	153.09	200.96		
14.72	1	637.24	195.08	161.72		

The format is:

- Column 1: the current time. Note that time starts at the time of arrival of the first vehicle;
- Column 2: The number of trucks currently on the bridge;
- Columns 3+: The current value of each load effect is given, according to the order of the load effects in the bridge definition file.

This file can get extremely large, but for short runs (e.g. 1-day) it is very useful for checking and debugging output.

An example output is given showing the number of trucks on the bridge through the day. As can be seen, four 3-truck events occur on this 20 m bridge.



All Events File

If all loading events are selected to be output (Line 26 of BTLSin.txt), BTLS creates a single file for each bridge named:

BL_*L*_AllEvents.txt

Where *L* is the bridge length. The file gives the maximum of each calculated load effect recorded during each loading event to occur. A loading event is defined as the loading that occurs between two occasions of zero trucks on the bridge, or the departure or arrival of another vehicle. A sample output is:

File	Edit	Format	View	Help
24.07	1	724.093	155.871	19.2673
30.5	1	157.071	29.5073	31.897
31.42	1	126.447	30.4906	12.6447
36.69	1	1048.15	208.132	210.562
38.01	1	271.711	107.71	27.1711
52.05	1	0	0	35.439
52.14	2	899.714	181.25	140.176
53.57	1	65.2051	26.1082	3.7653
75.82	1	560.757	125.211	7.60991
76.57	1	746.83	172.656	20.7378
93.79	1	496.82	111.131	13.8124
95.41	1	153.275	60.5421	8.23547
106.71	1	18.5711	9.00651	21.4258
106.81	1	287.122	55.864	57.509
118.24	1	0	0	0
118.27	1	563.555	121.984	15.8094
171.66	1	344.476	73.9569	6.92291
172.69	1	333.893	70.1047	9.9831

The format is:

- Column 1: the starting time of the loading event.
- Column 2: The number of trucks in the event;
- Columns 3+: The maximum value of each load effect during the event.

This file can get extremely large, but is useful for checking and debugging output.

Fatigue Events File

If fatigue events are selected to be output (Line 28 of BTLSin.txt), BTLS creates a single file for each bridge named:

BL_L_Fatigue.txt

Where L is the bridge length. The file gives the maximum and minimum values of each loading event (defined above) in chronological order. This is suitable for examination of fatigue cycles. A sample output is shown below:

File	Edit	Format	View	Help				
16.54			16.54	0.00	16.54	0.00	17.04	226.25
	1		17.34	624.72	17.24	188.50	17.74	12.16
42.08			42.08	0.00	42.08	0.00	42.58	179.75
	1		42.68	629.86	42.68	148.88	43.28	1.90
63.12			63.12	0.00	63.12	0.00	63.62	244.51
	1		63.82	793.23	63.72	224.91	64.22	5.16
88.34			88.34	0.00	88.34	0.00	88.64	249.51
	1		88.94	823.87	88.84	171.38	89.34	8.19
139.19			139.19	0.00	139.19	0.00	139.59	160.71
	1		139.79	472.26	139.79	117.89	140.29	7.43

The format is:

- Column 1 (Line 1): the starting time of the loading event;
- Column 2 (Line 2): The number of trucks in the event.

For each line, the subsequent columns are:

- Column 3: The time at which the value of load effect 1 is recorded;
- Column 4: the value of load effect 1.

Columns 5 & 6 give the time and value of load effect 2, and so on.

For example, this file shows that load effect 3 had a maximum value at 17.04 s, followed by a minimum at 17.74 s.

5.3 Vehicle Output

Traffic File

A traffic file can be output in any Program Mode if selected on Line 29 of BTLSin.txt. For Program Mode 2, this must be selected.

The file is named as specified on Line 31 of BTLSin.txt.

The program outputs vehicles in CASTOR format. An example of the program output for several trucks is given:

```
1001 1 1 2 0 12618155 54 43211 18 2743 27 0 0 0 0 0 0 0 0 0 0 0 0
1001 1 1 2 0 2 412133 137 67311 18 5441 6626 17 0 0 0 0 0 0 0 0 0 0
1001 1 1 2 0 2 598157 64 43211 18 3243 32 0 0 0 0 0 0 0 0 0 0 0
1001 1 1 2 0 44354134 336133511 18 7839 7040 6327 6327 63 0 0 0 0 0 0 0 0
1001 1 1 2 0 93062152 117 67311 18 3941 3926 39 0 0 0 0 0 0 0 0 0 0
1001 1 1 2 0101765131 97 44211 18 5344 44 0 0 0 0 0 0 0 0 0 0 0 0
```

Flow Statistics

For generated or read-in vehicle files, selecting this option will output files containing the flow rate and traffic stream composition. The files are named:

FlowData_*D*_*L*.txt

Where *D* is the direction number and *L* is the lane number corresponding to those in the Lane Flow Data definition file.

For each hour of the simulation, the following statistics are collected for each lane:

- The number of vehicles in the hour
- The number of trucks;
- The number of cars;
- The number of 2-axle, 3-axle, 4-axle and 5-axle trucks in the hour.

This output allows direct comparison with the input given in the Lane Flow Definition file is generating traffic. A screenshot is shown below:

Hour	No. Vehicles	No. Trucks	No. Cars	2-Axles	3-Axles	4-Axles	5-Axles
0	175	175	0	41	9	42	83
1	121	121	0	31	1	36	53
2	145	145	0	35	4	52	54
3	139	139	0	34	1	44	60
4	125	125	0	21	1	35	68
5	129	129	0	24	2	50	53
6	155	155	0	28	2	62	63

5.4 Block Maximum Files

If block maximum vehicle files are selected to be output (Line 34 of BTLSin.txt), BTLS can create different types of file output.

Output by Number of Trucks

This option is specified on Line 37 of BTLSin.txt. Output is given for each load effect of each bridge named:

BM_V_L_N.txt

Where *L* is the bridge length and *N* is the number of trucks comprising the loading events. In this manner a comprehensive breakdown of the causing of loading can be studied, suitable for application of the CDS method (Caprani et al 2008).

A sample output is shown:

```

1 1475.9 84395.5 17.58 2
1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0
1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0
2 409.0 84395.4 15.07 2
1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0
1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0
3 378.4 84395.3 12.56 2
1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0
1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0
2
1 1198.8 100764.2 26.65 2
1001 2 1 0 3592311240 449115511 18 683311857 8812 8813 88 0 0 0 0 0 0 0
1001 2 1 0 3592372235 609108511 18 9429187561091010912109 0 0 0 0 0 0 0
2 324.5 123199.1 -1.71 2
1001 2 1 0101318 9222 219 61222 18 7461145 0 0 0 0 0 0 0 0 0 0 0
1001 2 1 010131847242 600110511 1811030140581171211710117 0 0 0 0 0 0 0
3 426.2 100764.2 26.65 2
1001 2 1 0 3592311240 449115511 18 683311857 8812 8813 88 0 0 0 0 0 0 0
1001 2 1 0 3592372235 609108511 18 9429187561091010912109 0 0 0 0 0 0 0

```

This file structure is termed a Loading Event File, and its structure explained later.

Summary Files

If block maximum summary files are selected to be output (Line 38 of BTLSin.txt), BTLs creates a file for each load effect of each bridge named:

BM_S_L_Eff_E.txt

Where L is the bridge length and E is the load effect number. The file gives the maximum load effect recorded during each block, broken down according to the number of trucks comprising the event. A sample output is:

```

1 1350.8 1316.5 1009.9
2 1251.7 1275.2 1410.0
3 1345.4 1283.4 1195.6
4 1267.8 1296.1 1318.5
5 1353.6 1590.8 1014.6
6 1237.6 1153.9 1061.7
7 1225.0 1278.4 1148.6
8 1259.7 1227.1 1116.1
9 1376.4 1340.1 1146.4
10 1412.3 1325.8 1073.1

```

The format is:

- Column 1: The block index.
- Column 2: The block maximum 1-truck load effect;
- Columns 3: The block maximum 2-truck load effect
- Columns 4+: As appropriate, the block maximum 3-, 4-, ... truck load effect.

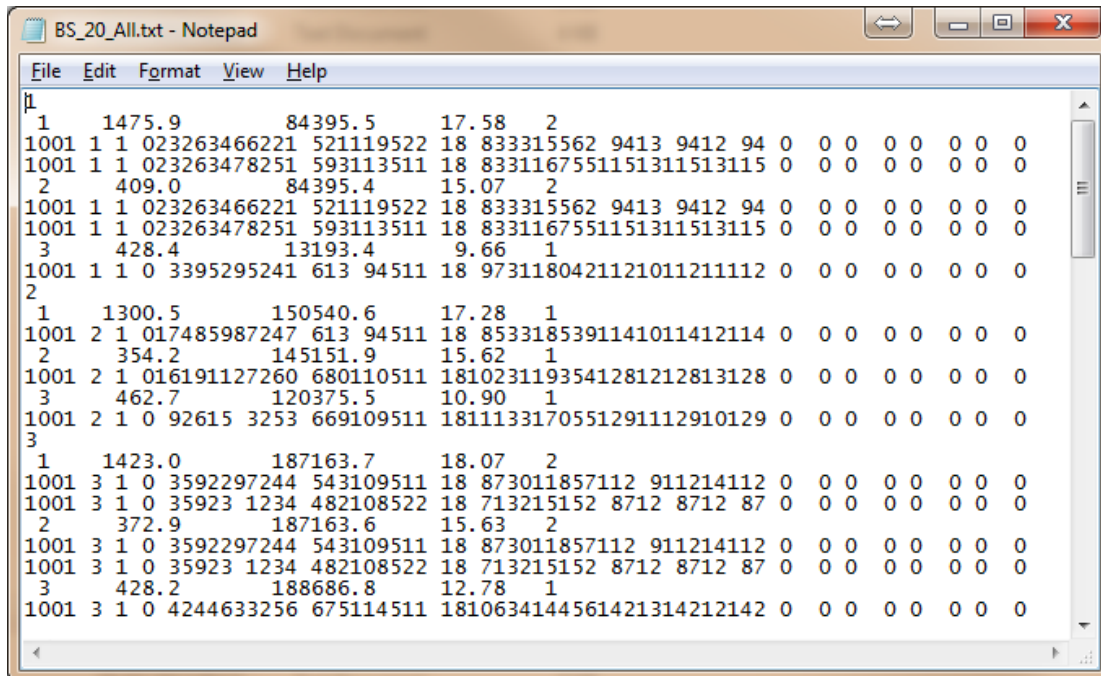
Taking the maximum across Columns 2+ gives the overall block maximum load effect.

Mixed Vehicle Output

This option is specified on Line 39 of BTLSin.txt. This form of output represents the conventional form in which the number of trucks comprising the event is not taken into account. Instead, the maximum load effect recorded during the block is noted. One such file per bridge is output, named:

BM_V_L_All.txt

Where L is the bridge length. This file structure is a Loading Event File, explained later. A sample output showing 3 blocks is:



As can be seen, there are a different number of trucks comprising the loading events that cause the maximum of each load effect in the block. Sometimes the same truck(s) loading event causes the maximum of 2 or more load effects, but in general different loading events cause the maximum of each load effect. This is because of the different shapes of the influence lines, and hence critical loading arrangements.

5.5 Peaks-Over-Threshold Files

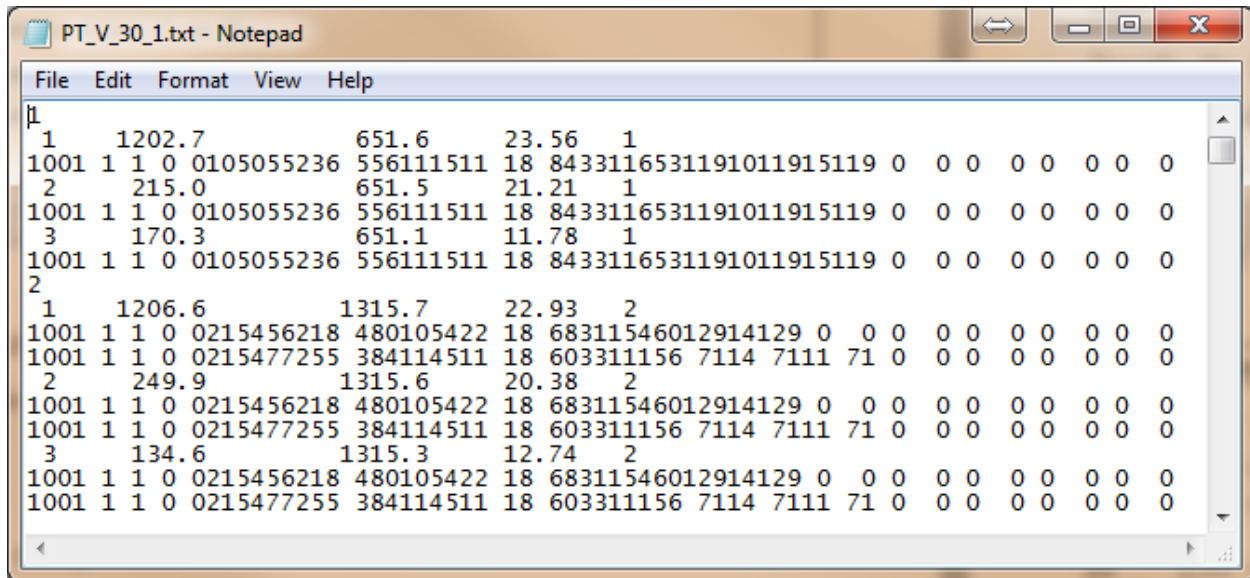
If POT files are selected to be output (Line 41 of BTLSin.txt), BTLS can create two types of file output.

Vehicle Files

This option is specified on Line 42 of BTLSin.txt. The vehicles comprising the loading event that is a peak are output in a Loading Event File structure. The files are named:

PT_V_L_E.txt

Where L is the bridge length and E is the load effect number. This file structure is a Loading Event File, explained later. A sample output showing 2 peaks is:



Note that the number of vehicles comprising the loading events are mixed and not separated out.

Summary Files

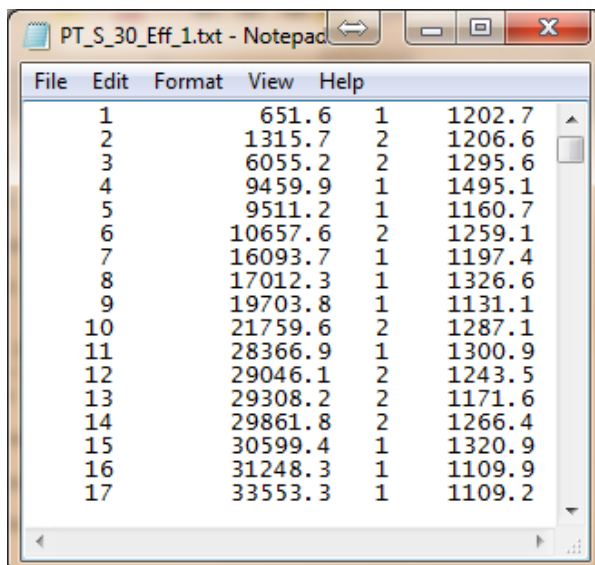
This option is specified on Line 43 of BTLSin.txt. The vehicles comprising the loading event that is a peak are output in a Loading Event File structure. The files are named:

PT_S_L_Eff_E.txt

Where L is the bridge length and E is the load effect number. Each row of data in this file corresponds to a recorded peak. For each peak, the following data is output:

- The peak number;
- The time at which the peak occurred;
- The number of truck sin the event;
- The peak load effect value.

A screenshot of such a file is shown below.



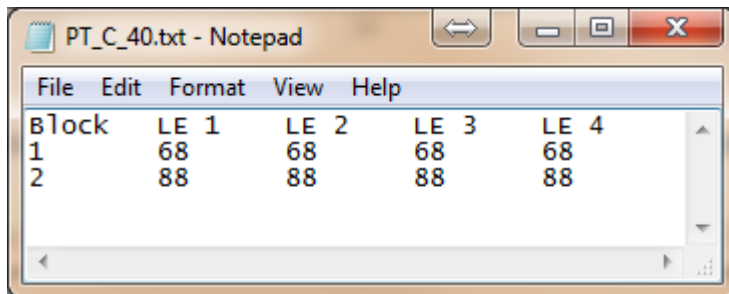
Note that since there is not much memory required to store peak information, the buffer size defined on Line 36 of BTLSin.txt should be large to prevent frequent disk writing which is slow.

Counter Files

This option is specified on Line 44 of BTLSin.txt. The number of peaks over the specified threshold occurring in each counter interval (defined on Lines 45-6 of BTLSin.txt) is output for each load effect. The files are named:

PT_C_L.txt

Where L is the bridge length. Each row of data in this file corresponds to a counter interval, and for each block and load effect the number of peaks is given. A screenshot of such a file is shown below.



Block	LE 1	LE 2	LE 3	LE 4
1	68	68	68	68
2	88	88	88	88

If the POT output buffer size (Line 47 of BTLSin.txt) is such that an output occurs during a counter interval, then the interval will have two rows (the number of peaks before output, and number after output) and the total number of peaks will be the sum of the two results. Again, since there is not much memory required to store peak information, the buffer size defined on should be large to prevent frequent disk writing which is slow.

5.6 Load Effect Statistics Output

BTLS can output some useful summary statistics of the calculated load effects. The statistics currently supported are:

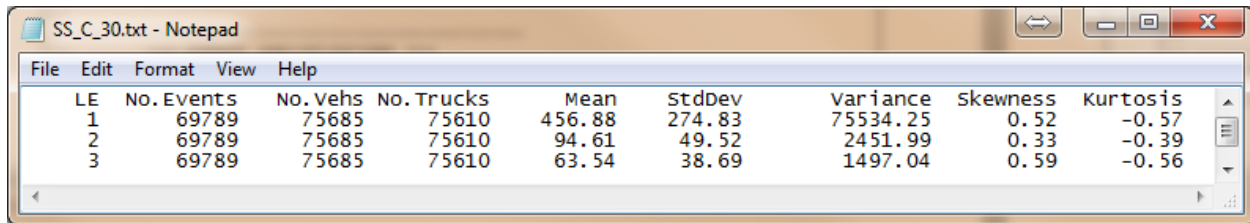
- Events count;
- The number of vehicles recorded;
- The number of trucks recorded
- The minimum load effect value;
- The maximum load effect value;
- The mean load effect value;
- The standard deviation of load effect;
- The load effect variance;
- The load effect skewness;
- The load effect kurtosis;
- The number of events comprising 1, 2, and on, trucks.

Cumulative Statistics

For this output, the statistics are accumulated throughout the full length of the simulation. The files are named:

SS_C_L.txt

Where L is the bridge length. In this file, each row corresponds to a load effect. A sample screenshot is:



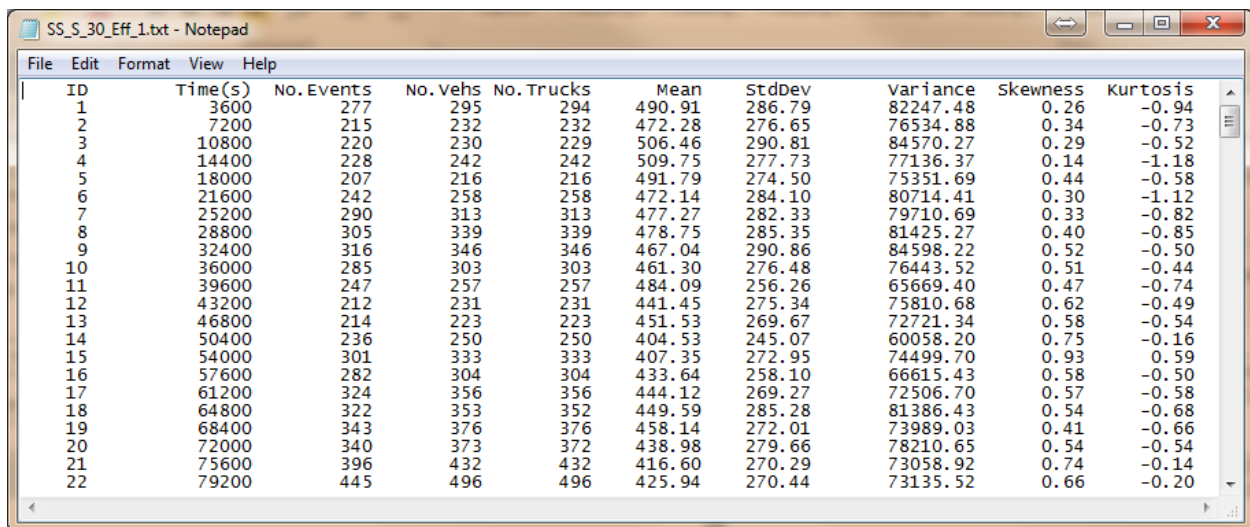
LE	No. Events	No. Vehs	No. Trucks	Mean	StdDev	Variance	Skewness	Kurtosis
1	69789	75685	75610	456.88	274.83	75534.25	0.52	-0.57
2	69789	75685	75610	94.61	49.52	2451.99	0.33	-0.39
3	69789	75685	75610	63.54	38.69	1497.04	0.59	-0.56

Interval Statistics

For intervals of specified duration (Line 51 of BTLSin.txt), the statistics are recorded and written to disk when the buffer size is exceeded (Line 41 of BTLSin.txt). This buffer size should be quite large since not much memory is needed to store statistics. The files are named:

SS_S_L_Eff_E.txt

Where L is the bridge length and E is the load effect number. A sample output is shown below. The interval number and time are also given.



ID	Time(s)	No. Events	No. Vehs	No. Trucks	Mean	StdDev	Variance	Skewness	Kurtosis
1	3600	277	295	294	490.91	286.79	82247.48	0.26	-0.94
2	7200	215	232	232	472.28	276.65	76534.88	0.34	-0.73
3	10800	220	230	229	506.46	290.81	84570.27	0.29	-0.52
4	14400	228	242	242	509.75	277.73	77136.37	0.14	-1.18
5	18000	207	216	216	491.79	274.50	75351.69	0.44	-0.58
6	21600	242	258	258	472.14	284.10	80714.41	0.30	-1.12
7	25200	290	313	313	477.27	282.33	79710.69	0.33	-0.82
8	28800	305	339	339	478.75	285.35	81425.27	0.40	-0.85
9	32400	316	346	346	467.04	290.86	84598.22	0.52	-0.50
10	36000	285	303	303	461.30	276.48	76443.52	0.51	-0.44
11	39600	247	257	257	484.09	256.26	65669.40	0.47	-0.74
12	43200	212	231	231	441.45	275.34	75810.68	0.62	-0.49
13	46800	214	223	223	451.53	269.67	72721.34	0.58	-0.54
14	50400	236	250	250	404.53	245.07	60058.20	0.75	-0.16
15	54000	301	333	333	407.35	272.95	74499.70	0.93	0.59
16	57600	282	304	304	433.64	258.10	66615.43	0.58	-0.50
17	61200	324	356	356	444.12	269.27	72506.70	0.57	-0.58
18	64800	322	353	352	449.59	285.28	81386.43	0.54	-0.68
19	68400	343	376	376	458.14	272.01	73989.03	0.41	-0.66
20	72000	340	373	372	438.98	279.66	78210.65	0.54	-0.54
21	75600	396	432	432	416.60	270.29	73058.92	0.74	-0.14
22	79200	445	496	496	425.94	270.44	73135.52	0.66	-0.20

5.7 Loading Event File Structure

This file structure contains all information relating to the event. An example is shown below (from the Block Maximum Vehicle Files, Out by Number of Trucks screenshot).

Line	Data
1	1
2	1 1475.9 84395.5 17.58 2
3	1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0 0
4	1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0 0
5	2 409.0 84395.4 15.07 2
6	1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0 0
7	1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0 0
8	3 378.4 84395.3 12.56 2
9	1001 1 1 023263466221 521119522 18 833315562 9413 9412 94 0 0 0 0 0 0 0 0
10	1001 1 1 023263478251 593113511 18 8331167551151311513115 0 0 0 0 0 0 0 0

11	2
12	1 1198.8 100764.2 26.65 2
13	1001 2 1 0 3592311240 449115511 18 683311857 8812 8813 88 0 0 0 0 0 0 0
14	1001 2 1 0 3592372235 609108511 18 9429187561091010912109 0 0 0 0 0 0 0
15	2 324.5 123199.1 -1.71 2
16	1001 2 1 0101318 9222 219 61222 18 7461145 0 0 0 0 0 0 0 0 0 0 0
17	1001 2 1 010131847242 600110511 1811030140581171211710117 0 0 0 0 0 0 0
18	3 426.2 100764.2 26.65 2
19	1001 2 1 0 3592311240 449115511 18 683311857 8812 8813 88 0 0 0 0 0 0 0
20	1001 2 1 0 3592372235 609108511 18 9429187561091010912109 0 0 0 0 0 0 0

Each line is explained as follows.

Line 1: The index of the current block (if block maximum output), or the index of the particular loading event in legacy files.

Line 2: This is the load effect information line with 5 fields of data separated by tabs:

- Field 1: The load effect number;
- Field 2: The value of the load effect;
- Field 3: The time at which this load effect was found in seconds;
- Field 4: The distance of the first axle of the first truck on the bridge relative to the bridge datum, at the time of the crossing event maximum effect being reached. This allows one to sketch the positions of the trucks at the time of the load effect.
- Field 5: The number of trucks comprising the event.

Line 3-4: These lines provide the truck data string in CASTOR format for later processing.

Line 5+: The format of lines 2-4 continues for each of the effects calculated. Line 11 then provides the information for the start of the second block or loading event, and the format repeats itself.

6 Appendices

6.1 Appendix 1 – Traffic File Formats

CASTOR File Format

In the table below, the Format column gives the storage type of the data. IX refers to an integer of X number of digits, including leading or trailing zeros.

Record	Unit	Format
Head		I4
Day		I2
Month		I2
Year		I2
Hour		I2
Minute		I2
Second		I2
Second/100		I2
Speed	dm/s	I3

Record	Unit	Format
Gross Vehicle Weight - GVW	kg/100	I4
Length	dm	I3
Number of Axles		I1
Direction (1 or 2)		I1
Lane		I1
Transverse Location In Lane	dm	I3
Weight Axle 1	kg/100	I3
Spacing Axle 1 - Axle 2	dm	I2
Weight Axle 2	kg/100	I3
Spacing Axle 2 - Axle 3	dm	I2
⋮	⋮	⋮
Spacing Axle 8 - Axle 9	dm	I2
Weight Axle 9	kg/100	I3

BeDIT File Format

This file format is similar to CASTOR except that the maximum number of axles possible is 20, the axle spacings are given by a three digit number, and the direction is zero-based.

Record	Unit	Format
Head		I4
Day		I2
Month		I2
Year		I2
Hour		I2
Minute		I2
Second		I2
Second/100		I2
Speed	dm/s	I3
Gross Vehicle Weight - GVW	kg/100	I4
Length	dm	I3
Number of Axles		I2
Direction (zero-based)		I1
Lane		I1
Transverse Location In Lane	dm	I3
Weight Axle 1	kg/100	I3
Spacing Axle 1 - Axle 2	dm	I3
Weight Axle 2	kg/100	I3
Spacing Axle 2 - Axle 3	dm	I3
⋮	⋮	⋮
Spacing Axle 19 - Axle 20	dm	I3
Weight Axle 20	kg/100	I3

DITIS File Format

This file format is similar to BeDIT except that the wheel track width (i.e. transverse width of the vehicle, wheel-to-wheel) is included for each axle; the transverse position units are cm; and the direction is one-based. This is for use with influence surfaces. Further, the year is a four-digit number.

Field	Record	Unit	Format	Start
1	Head		I4	1
2	Day		I2	5
3	Month		I2	7
4	Year		I4	9
5	Hour		I2	13
6	Minute		I2	15
7	Second		I2	17
8	Second/100		I2	19
9	Speed	dm/s	I3	21
10	Gross Vehicle Weight - GVW	kg/100	I4	24
11	Length	dm	I3	28
12	Number of Axles		I2	31
13	Direction (1 or 2)		I1	33
14	Lane		I1	34
15	Transverse Location In Lane	cm	I3	35
16	Weight Axle 1	kg/100	I3	38
17	Track Width Axle 1	cm	I3	41
18	Spacing Axle 1 - Axle 2	dm	I3	44
19	Weight Axle 2	kg/100	I3	47
20	Track Width Axle 2	cm	I3	50
21	Spacing Axle 2 - Axle 3	dm	I3	53
⋮	⋮	⋮	⋮	⋮
72	Spacing Axle 19 - Axle 20	dm	I3	206
73	Weight Axle 20	kg/100	I3	209
74	Track Width Axle 20	cm	I3	212

MON File Format

This file format is roughly similar to the BeDIT format but the units, order, and number of digits in each field varies to match common Australian WIM measurements. There is no defined maximum number of axles, but the last axle spacing is defined, allowing calculation (from the vehicle length) of the front and rear overhangs.

Record	Unit	Format
Head		I9
Day		I2
Month		I2
Year		I4
Hour		I2
Minute		I2
Second	ms	I5
Number of axles		I2
Number of axle groups		I2
Gross Vehicle Weight - GVW	kg	I6
Speed	km/h	I3
Length	mm	I5
Lane (1-based local)		I1
Direction (0-based)		I1
Transverse Location In Lane	mm	I4
Weight Axle 1	kg	I5
Spacing Axle 1 - Axle 2	mm	I5

Record	Unit	Format
Weight Axle 2	kg	I5
Spacing Axle 2 - Axle 3	mm	I5
⋮	⋮	⋮
Weight Axle – Maximum defined	kg	I5
Spacing Axle – Maximum defined	mm	I5

SiWIM CSV File Format

Introduced in v1.3.7, this format (format code 5) reads the CSV export of the SiWIM[®] bridge weigh-in-motion system. Unlike the fixed-width formats above, it is a comma-separated file whose first line is a header row; columns are located by header name, so the column order is not fixed and unrecognised columns are ignored. This format is **read-only**: it cannot be selected as the vehicle output file format.

The columns used by BTLS are:

Header name	Unit	Description
wim.ts		Timestamp, YYYY-MM-DD-HH-MM-SS-mmm (millisecond resolution)
wim.lane		Lane number, 1-based local (defaults to 1 if absent); all vehicles are assigned direction 1
wim.v	m/s	Vehicle velocity
wim.naxles		Number of axles (must be at least 1)
wim.axgrps		Axle group pattern, e.g. 1233 for a 9-axle vehicle in 4 groups; the number of groups is taken as the number of digits. If absent or invalid, groups are inferred from the axle spacings (a spacing greater than 2.0 m starts a new group)
wim.gvw	kN	Gross vehicle weight
wim.whlbse	m	Wheelbase; used as the vehicle length
wim.acws.w.i	kN	Weight of axle i (0-based), for $i = 0 \dots \text{naxles} - 1$
wim.ads.d.i	m	Spacing between axle i and axle $i + 1$ (0-based)

The transverse position in lane is not present in the SiWIM export and is set to zero. The first line of the file must be the header row; a file without a header, a bad timestamp, or a row with an invalid axle count is reported as an input error.

6.2 Appendix 2 – References

- Caprani, C.C. (2005), *Probabilistic Analysis of Highway Bridge Traffic Loading*, PhD Thesis, School of Architecture, Landscape and Civil Engineering, University College Dublin, Ireland. <http://www.colincaprani.com/research/phd-dissertation/>
- Caprani, C.C. (2012), ‘Calibration of a congestion load model for highway bridges using traffic microsimulation’, *Structural Engineering International*, 22(3), August, in print.
- Caprani, C.C., O'Brien, E.J. and McLachlan, G.J. (2008), ‘Characteristic traffic load effects from a mixture of loading events on short to medium span bridges’, *Structural Safety*, Vol. 30(5), September, pp. 394-404.
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- Grave, S.A.J. (2001), *Modelling of Site-Specific Traffic Loading on Short to Medium Span Bridges*, Ph.D. Thesis, Department of Civil Engineering, Trinity College Dublin.

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